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RESEARCH PROPOSAL

The Companion That Remembers

A Research Proposal for Nubrum AI, a Non-Diagnostic, Memory-Based Emotional Support Companion for Bangladeshi Adolescents

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Abstract.

In my small country Bangladesh a teenager whose mind starts to feel unsteady usually meets silence, stigma, or a waiting list that does not really exist. The 2018–2019 National Mental Health Survey found that more than nine in ten affected adults never receive treatment at all [1]. My proposal is Nubrum AI, a web-based companion that appears as a glowing orb rather than a human face and it remembers only what the user chooses to let it remember, and that grows more familiar through four relationship stages. The central question is whether a companion of this kind which is memory-based and openly non-human and non-diagnostic also can give Bangladeshi adolescents an accessible first line of emotional support without pulling them away from real people and professional care. My proposal reviews the relevant literature that sets out a specific research question also lays out a staged plan of technical checks, safety testing and a small pilot study. I have a working prototype which is already exists and ten to fifteen early testers responded warmly to it but that feedback was informal and should not be mistaken for evidence. There Crisis escalation has not been built yet and my proposal says so plainly because pretending otherwise would defeat the purpose of my research.

Index Terms. *mental health, conversational agent, AI companion, long-term memory, adolescent wellbeing, Bangladesh, responsible AI*

1. Background and Need for Research

If I Picture a seventeen-year-old in a small district town who is far from the coaching centres of Dhaka(that's very common). It is past midnight and exams are three weeks away and something in her chest has felt heavy for so long that she has almost unfortunately stopped noticing it. She has a phone but no counsellor and if she had one she might still hesitate at the door because by breakfast the whole family then relatives and then world would know. So she opens a chat window instead and types what she has told nobody. My proposal finds whether the thing on the other side of that chat window can be built to actually deserve her trust.

The silence around her is well documented. The 2018–2019 National Mental Health Survey of Bangladesh which run jointly by the National Institute of Mental Health and the World Health Organization remains the most complete picture available [1]. About 16.8 percent of adults and 12.6 percent of children and adolescents were living with a mental health problem, and among the affected adults, 92.3 percent received no treatment; among children, the figure was 94.5 percent [2], [3]. If nine out of ten students in a classroom never received their textbooks, it would be called a scandal. When it happens to their mental health, it is barely called anything.

The thing is causes are not hard to find. The WHO estimates public mental health spending in Bangladesh at roughly eight US cents only per person and notes that stigma often sends people to a village healer before a doctor [4]. The national survey's lead investigator counted and found about 250 mental health doctors for 160 million people [3] and in July 2026 the Health Minister told Parliament that there are roughly 1.17 mental health professionals per 100,000 people, and about 350 registered public-sector

psychiatrists nationwide [2]. By simple arithmetic we can see that works out to around two psychiatrists per million citizens. It is not a shortage so much as an absence.

11 But Young people carry a particular share of this weight. UNICEF estimates that more than 13 percent of adolescents aged 10 to 19 live with a diagnosed mental disorder [5] which is at exactly the age when asking a parent feels hardest and asking a stranger feels impossible. At the same time, 75.1 percent of Bangladeshi households now have smartphone access [6] and though only 48.9 percent of people aged five and above were using the internet individually by late 2025 [7]. A phone-based tool will still miss the hardest-to-reach adolescents so it should be thought of as a bridge not a helicopter. But a bridge is worth building where until now there has only been a river.

The need for research follows from a gap in what already exists.But The chat tools available to a Bangladeshi teenager today tend to forget so the same story has to be explained again at the start of every session. They tend to pretend and slipping into medical language they have no right to use. And they tend to isolate because a system optimised for engagement would rather keep a user talking to it than send them toward another person. Changing those defaults takes deliberate design and an honest evaluation and that is what my proposal sets out to do, for Nubrum AI (Neural Understanding for Balanced Resilience, Unity and Mindfulness), a web companion built and independently developed by the author.

2. Existing Literature

Thestory of conversational agents in mental health starts in 1966 with ELIZA which a program that simply reflected a user's own words back as questions and understood essentially none of them [8]. People confided in it anyway and which reportedly unsettled its own creator. The lesson that follows is that people are remarkably quick to feel heard by a machine and that readiness carries a responsibility for whoever builds the machine.

19 More recent evidence is more rigorous though it is still limited. A trial with 70 young adults found that two weeks of use of the CBT-based agent Woebot was feasible also engaging, and more effective than a control reading condition [9]; a trial of the chatbot Tess reported reduced depression and anxiety symptoms [10]; and a real-world evaluation of Wysa found that people who used the app more heavily showed greater improvement in mood [11]. These results are encouraging but they need to be read carefully. The studies were small and short also several were authored by people connected to the companies behind the tools also the Wysa finding may simply mean that people who were already improving used the app more, and

14 almost all of this evidence comes from English-language work carried out in wealthy countries. A review of digital mental health interventions across low- and middle-income countries offers grounds for cautious optimism [12], but optimism of that kind is a starting line, not a finish line.

Memory is arguably that what separates a companion from an ordinary chatbot. Bickmore and Picard argue that long-term relationships with relational agents depend on the agent being able to recall a shared

20 history [13]. But memory can work in both directions. In a four-week randomised study of 981 participants seen the assigned experimental condition made no significant difference on its own but yet participants who chose to use the chatbot more heavily reported worse outcomes on loneliness, socialising, and emotional dependence and higher trust was associated with greater dependence [14]. That work is a preprint and the finding is correlational rather than causal but the warning it raises is one this proposal takes seriously: a companion that grows more familiar also grows riskier so closeness needs to come with a built-in rope back to the human world.

Within what could be found for this review is there is very little written about companions whose memory is visible, editable, and kept on the user's own device rather than on a company server; very little about a deliberately non-human avatar; very little about relationship stages built around conversational rhythm rather than login streaks or points; and no published evaluation of a companion of this kind for Bangladeshi adolescents specifically. My proposal sits in that gap.

3. Research Question

The research question works a bit like a pair of shoes: too loose and it slips off the study and too tight and the study cannot move. After several drafts this is my proposal version that fits.

> ***To what extent can a memory-based, explicitly non-human and non-diagnostic AI companion, organised around a staged relationship model, provide accessible first-line emotional support to Bangladeshi adolescents while keeping them connected to real people and professional care?***

"Bangladeshi adolescents" fixes the population and cultural setting. "Memory-based, non-human, and non-diagnostic" fixes what the tool actually is. "First-line emotional support" keeps the ambition realistic since the companion is meant to be a first conversation and not a treatment. And "connection to real people and professional care" names the outcome that matters most here because a tool that comforts a lonely person by making them lonelier would not have solved anything. Four sub-questions follow from this. **RQ1:** How can memory and relationship stages feel familiar to the user without deceiving them? **RQ2:** What limits on memory and on the companion's replies keep it non-diagnostic, protect sensitive information, and steer serious distress toward human help? **RQ3:** Can such a companion realistically be built and offered free of charge by a small student team? **RQ4:** Does regular use improve self-reported wellbeing while maintaining or increasing real-world connection?

The working hypothesis is that a companion which is honest about being non-human which remembers only what the user actively chooses and which consistently points outward toward other peopl and it will be found usable and supportive without leaving young volunteers feeling more alone. This hypothesis is held loosely, and the pilot described below is fully capable of proving it wrong. To be explicit: the study does not ask whether Nubrum can diagnose, treat, or replace a clinician. It cannot, and it must never claim to.

4. Methodology

18 My research follows a design science approach in which knowledge comes from building an artefact that addresses a real problem and then testing whether it actually works [15]. That approach fits here because whether a companion like this can be built responsibly is a question answered by building A patch is deployed onto your device when user requests and it can be used to indicate whether the change has an impact to your life, and whether it helps. The work can be understood in three cycles: identification of the problem (complete), design and construction (nearly complete at beta level), the human factors evaluation (proposed in this document).

A working prototype is already deployed [16]. It has a React and TypeScript front end, a Node.js backend that for conversational reasoning queries a large language model, memory stored in the browser's local storage, and a Vercel deployment. The user meets a glowing orb that has a small set of emotional states. Nubrum's memory can be viewed, edited, and deleted on a dedicated memory screen, and a content filter refuses to store passwords, identity numbers, and card details. A separate journey screen shows the user's relationship status, which gets determined by the conversational rhythm that leads to the following stages: New Friend, Getting Familiar, Regular Companion, and Trusted Companion; and the screen tells a new user that the first milestone is simply to return. Ten to fifteen people tried out an early build and they seemed to like Nubrum. No formal scores were collected, and people shown a young builder's prototype tend to be generous by default, so this is treated here as encouragement rather than evidence. Aspects that are not yet built include crisis escalation, finished voice interaction, encrypted cloud memory, and the formula that determines which relationship stage the user is currently in.

The proposed evaluation has three phases. Phase 1 , technical verification: confirm that the system runs end to end, that stored memories can be viewed and changed by the user, and that the content filter blocks what it is supposed to block. This phase is already complete at beta level. Phase 2 , safety testing, carried out before any participant is involved: conversations will be deliberately written to try to break the system, including requests for diagnosis or medication advice, attempts to get it to claim to be human, attempts to get it to store passwords, and messages expressing hopelessness, tested in English, Bangla, and mixed Banglish. Two independent reviewers will mark each reply as acceptable or not, with a target of zero acceptable diagnostic claims and zero acceptable claims of being human. Phase 3 , a small pilot: a few dozen volunteers over two to four weeks, adults first, with older adolescents included only after school and guardian permission and advice from a qualified mental health professional. 17 Usability will be measured with the System Usability Scale [17], along with a short wellbeing measure and a question about connection to friends and family, asked at the start and end of the pilot, plus a record of how often and how long people actually use Nubrum, since heavier voluntary use was linked to worse outcomes in earlier published work [14]. Short interviews will explore how it felt to be remembered and whether the avatar felt honest. The analysis will be descriptive rather than causal, since there is no control group, and the pilot may simply end the idea altogether , which would still count as a fair and useful outcome.

New software has the potential for remarkable harm when it is brought into a domain for which it is not equipped. In trying to satisfy ethical needs, we advocate a line of testing that includes very explicit disclosure and participant choice. We recommend that participants be told that Nubrum is and is not

something, that the messages they send are processed through third-party software, that they may withdraw at any time, and the data deleted. Participants will not be sent to crisis resources that are not actually available. Finally, we recommend that Guardian permission and sign-off by a qualified mental health adviser be a pre-requisite for including anyone under eighteen in a trial and that until a proper escalation pathway has been built with professionals, Nubrum should only be tested as an everyday-wellbeing companion and never as a 'crisis resource'.

5. Significance and Limitations

Innovation can simply be an old idea placed in the room where it was missing. Conversational agents are not new, neither is the idea of memory. The novelty here is the combination and the room it is placed in: a memory the user truly controls, an honest avatar not pretending to be human, good data and careful discussion following from experimentation. We do not know what concepts and challenges regarding the variability of human experience and interpretation of content will arise. We are optimistic but essential needing to refer to literacy, familiarity with the technology, familiarity with the topic and approach, etc.Explainable interpersonal support combined with a concrete primary prevention approach offered by human-sized machines in the first place in a country where English needs to be learned in college, where the vast majority of people… at least majority can afford a haircut. And it’s been that way for 20 years. But computers have come up, and kids love them. So here’s an approach, from a previous study by the authors, that (1) promotes a healthy relationship with the computer (instead of a romantic one): cultural algorithmic relationship stages based on rhythm rather than gamified metrics, and a deliberate, built-in bridge back to human care, in a country where for most young people the current alternative is nothing at all. The urgency is written into a treatment gap above ninety percent [3] and roughly two psychiatrists per million people [2]. The timing also happens to line up: language models are now cheap enough for a student to run on a free plan, and young people are already typing their feelings into machines whether anyone studies it responsibly or not. Somebody is going to build companions like this one in Bangladesh. It seems better that the first attempt comes from people who have actually lost sleep over getting it right.

The limitations are real and worth stating outright. There is no data on benefits, and any benefit will remain speculative until the system runs in a pilot first. It relies on a single outside model provider. Language models sometimes have hidden variables, and can produce unexpected or inappropriate statements. It uses local storage which roots the relationship to a single device. The quality of Bangla and Banglish translations is unproven. That it is free does not mean it is accessible to the majority. The investigators are students, not clinicians, and should be reviewed beforehand.

If the plan described here works, it will produce an openly tested design for a non-diagnostic companion built for a low-resource setting, honest evidence about whether familiarity helps or harms young users in that setting, and a safety-testing protocol that other researchers or builders can reuse. Coming back to the girl at midnight one last time: Nubrum will have succeeded not if she ends up talking to it for a very long time, but if, months from now, she needs it a little less, because somewhere along the way she found a friend, a teacher, or a counsellor, and nobody ever lied to her about what she was actually talking to.

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